

1) Sử dụng cơ sở dữ liệu lưu trữ thông tin sinh viên, lớp, môn học được mô hình hóa bởi collection sinhvien. Thực hiện các yêu cầu sau:

a/ Viết lệnh thêm vào collection sinhvien trong 2 trường hợp: thêm một và

nhiều document.

The screenshot shows the MongoDB Compass web interface. On the left, a sidebar lists the database 'Buoi2' and its collections: 'sinhvien', 'admin', 'config', and 'local'. The 'sinhvien' collection is selected. The main area shows the 'Documents' tab with 0 documents. Below this, there are buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'. The document list is empty, but the schema shows fields: '_id: ObjectId', 'Masv: String', and 'Hoten: String'. The top bar shows the connection to 'localhost:27017' and the database 'Buoi2'.

```
>_MONGOSH
>use sinhvien; insert([{"Masv": "sv002", "Hoten": "Trần Minh"}, {"Masv": "sv003", "Hoten": "Phạm Tuấn"}])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8559b03e03deaab7ed4918'),
    '1': ObjectId('6a8559b03e03deaab7ed4919')
  }
}
```

localhost:27017

My Queries

Performance

Databases

Search

Buoi2

sinhvien

admin

config

local

My Queries

sinhvien

Buoi2

localhost:27017 > Buoi2 > sinhvien

Documents 0

Aggregations

Schema

Indexes 1

Validation

⌚

Type a query: { field: 'value' } or [Generate query](#)

+ ADD DATA

EXPORT DATA

UPDATE

DELETE

Masv : "sv002"

Hoten : "Trần Minh"

_id: ObjectId('6a8559b03e03deaab7ed4919')

Masv : "sv003"

Hoten : "Phạm Tuấn"

_id: ObjectId('6a855d083e03deaab7ed491a')

Masv : "sv004"

Hoten : "Trần Minh Nghĩa"

Ngaysinh : 2001-04-14T00:00:00.000+00:00

▸ Lop : Object

▸ Monhoc : Array (3)

```
>_MONGOSH
}
})
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a855d083e03deaab7ed491a')
  }
}
Buoi2>
```

b/ Viết lệnh xóa document với điều kiện mã sinh viên là “sv005”

The screenshot displays the MongoDB Compass interface and a terminal window. The Compass interface shows the 'sinhvien' collection in the 'Bui2' database. The 'Documents' tab is active, showing three documents. The first document has 'Masv: "sv002"' and 'Hoten: "Trần Minh"'. The second document has '_id: ObjectId("6a8559b03e03deaab7ed4919")', 'Masv: "sv003"', and 'Hoten: "Phạm Tuấn"'. The third document has '_id: ObjectId("6a855d083e03deaab7ed491a")', 'Masv: "sv004"', 'Hoten: "Trần Minh Nghĩa"', 'Ngaysinh: 2001-04-14T00:00:00.000+00:00', 'Lop: Object', and 'Monhoc: Array (3)'. The terminal window shows the command 'db.sinhvien.deleteOne({ Masv: "sv005" })' being executed, resulting in 'acknowledged: true' and 'deletedCount: 1'.

localhost:27017 ...

My Queries Performance Databases Search Buoi2 sinhvien admin config local

localhost:27017 > Buoi2 > sinhvien

Documents 0 Aggregations Schema Indexes 1

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA EXPORT DATA UPDATE DELETE

Masv : "sv002"
Hoten : "Trần Minh"

_id: ObjectId('6a8559b03e03deaab7ed4919')
Masv : "sv003"
Hoten : "Phạm Tuấn"

_id: ObjectId('6a855d083e03deaab7ed491a')
Masv : "sv004"
Hoten : "Trần Minh Nghĩa"
Ngaysinh : 2001-04-14T00:00:00.000+00:00
Lop : Object
Monhoc : Array (3)

```
>_MONGOSH
{
  acknowledged: true,
  deletedCount: 0
}
> db.sinhvien.deleteOne({ Masv: "sv005" })
< {
  acknowledged: true,
  deletedCount: 1
}
Buoi2>
```

c/ Viết lệnh xóa những sinh viên học lớp có mã lớp là 103

The image shows a MongoDB Compass interface and a terminal window. The Compass interface is connected to `localhost:27017` and shows the `Buoi2` database with the `sinhvien` collection selected. The `Documents` tab is active, showing three documents. The first document has `Masv: "sv002"` and `Hoten: "Trần Minh"`. The second document has `_id: ObjectId('6a8559b03e03deaab7ed4919')`, `Masv: "sv003"`, and `Hoten: "Phạm Tuấn"`. The third document has `_id: ObjectId('6a855d083e03deaab7ed491a')`, `Masv: "sv004"`, `Hoten: "Trần Minh Nghĩa"`, `Ngaysinh: 2001-04-14T00:00:00.000+00:00`, `Lop: Object`, and `Monhoc: Array (3)`.

The terminal window shows the following commands and output:

```
> _MONGOSH
acknowledged: true,
  deletedCount: 1
}
> db.sinhvien.deleteMany({ "Lop.MaLop": "103" })
< {
  acknowledged: true,
  deletedCount: 0
}
Buoi2>
```

d/ Viết lệnh sửa Họ tên của sinh viên có mã sv001 thành Đỗ Nhật Lâm

The screenshot displays the MongoDB Compass interface at localhost:27017, connected to the 'Bui2' database and the 'sinhvien' collection. The 'Documents' tab shows three records. The first record, with '_id' ObjectId('6a8559443e03deaab7ed4915'), has 'Masv' 'sv001' and 'Hoten' 'Đỗ Nhật Lâm'. Below this, a terminal window shows the output of a MongoDB update command:

```
>_MONGOSH
/
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

The terminal prompt is 'Bui2>'. The Compass interface also shows a search bar and buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'.

e/ Sửa tuổi thành 25, Phái thành Nữ, Họ tên thành Trần Thị Lan cho sinh viên

có mã là sv003

localhost:27017

My Queries

Performance

Databases

Search

Buoi2

sinhvien

admin

config

local

My Queries

sinhvien

Buoi2

+

localhost:27017 > Buoi2 > sinhvien

Documents 0

Aggregations

Schema

Indexes 1

Vali

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

EXPORT DATA

UPDATE

DELETE

_id: ObjectId('6a8559b03e03deaab7ed4919')

Masv : "sv003"

Hoten : "Trần Thị Lan"

Phai : "Nu"

Tuoi : 25

_id: ObjectId('6a855d083e03deaab7ed491a')

Masv : "sv004"

Hoten : "Trần Minh Nghĩa"

Ngaysinh : 2001-04-14T00:00:00.000+00:00

▶ Lop : Object

▶ Monhoc : Array (3)

>_MONGOSH

< {
 acknowledged: true,
 insertedId: null,
 matchedCount: 1,
 modifiedCount: 1,
 upsertedCount: 0
}

Buoi2>

f/ Sửa ngoại ngữ thứ 2 của sinh viên có mã sv003 thành Tiếng Hàn

localhost:27017 > Buo2 > sinhvien

Documents 0 Aggregations Schema Indexes 1 Validate

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA EXPORT DATA UPDATE DELETE

Document 1:

- _id: ObjectId('6a8559b03e03deaab7ed4919')
- Masv: "sv003"
- Hoten: "Tran Thi Lan"
- Phai: "Nu"
- Tuoi: 25
- Ngoaingu: Object
 - 1: "Tiếng Hàn"

Document 2:

- _id: ObjectId('6a855d083e03deaab7ed491a')
- Masv: "sv004"
- Hoten: "Trần Minh Nghĩa"
- Ngaysinh: 2001-04-14T00:00:00.000+00:00
- Lop: Object
- Monhoc: Array (3)

```
>_MONGOSH
/
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```


g/ Sửa điểm của sinh viên có mã sv003 học môn thứ 1 thành 9

The screenshot displays a MongoDB management interface. On the left, the 'Databases' sidebar shows a collection named 'sinhvien' under the 'Buoi2' database. The main area on the right shows a JSON document for a student with ID 'm001'. The document structure is as follows:

```
{
  "Mamh": "m001",
  "Tenmh": "Cơ sở dữ liệu",
  "Sotc": 3,
  "Diem": 9,
  "Monhoc": [
    {
      "Mamh": "m002",
      "Tenmh": "Toán cao cấp",
      "Sotc": 2,
      "Diem": 6.5
    },
    {
      "Mamh": "m003",
      "Tenmh": "Tiếng Anh",
      "Sotc": 1,
      "Diem": 8.5
    }
  ],
  "Ngoaingu": [
    {
      "Mamh": "m004",
      "Tenmh": "Tiếng Pháp",
      "Sotc": 1,
      "Diem": 7.5
    }
  ],
  "Tuoi": 25
}
```

Below the document, the MongoDB command and its output are shown in a terminal window:

```
>_MONGOSH
>
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

h/ Viết lệnh thay thế một document với `_id` được chỉ định.

The image shows a MongoDB Compass interface and a terminal window. The Compass interface is connected to `localhost:27017` and shows the `Buoi2` database with the `sinhvien` collection selected. The `Documents` tab is active, showing 3 documents. The first document is partially visible, showing `Ngaysinh` and `Lop`. The second document is fully visible, showing `_id`, `Masv`, `Hoten`, `Ngaysinh`, `Phai`, `Email`, `Lop`, `Monhoc`, `Ngoaingu`, and `Tuoi`.

The terminal window shows the following command and output:

```
>_MONGOSH
>
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
Buoi2>
```